

Divya Jyoti

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Professional Summary

- AI and Data Science professional with experience in Python, Machine Learning, Generative AI, SQL, AWS, and data analytics.
- Skilled in building intelligent applications using LLMs, Retrieval-Augmented Generation (RAG), LangChain, FastAPI, vector databases, and cloud technologies. Strong foundation in statistical modeling, NLP, deep learning, and data visualization.
- Python-based data processing, and dashboard development, now focused on applied data science for marketing intelligence. Experienced in translating data into actionable business insights for stakeholders across banking and operations. Hands-on with NLP text analysis, clustering techniques, LLM API integration (Claude/Open AI), and building data pipelines that surface content-level insights. Eager to own and advance the modeling side of a marketing insights platform.

Technical Skills

- **Libraries:** Claude API, Open AI API, LangChain, Keras, PyTorch, Pandas, NumPy, Scikit-learn, Spicy, NetworkX, OpenCV, GNLTK, Matplotlib, chromadb, psycopg2, Ragas, fastAPI.
- **Programming Languages:** Python, SQL.
- **ML/AI:** GenAI, Classical ML (Supervised and Unsupervised), Graph Data Mining and Embeddings, Representation Learning, Reinforcement Learning, Natural Language Processing, Deep Learning (CNN,RNN), Computer Vision.
- **Database:** MySQL, SQL Server.
- **Cloud Computing:** Amazon Web Services (AWS).
- **Business Intelligence:** Power BI, Tableau, KPI dashboard
- **Enterprise Skills:** GIT, GitHub.
- **IDE:** Jupyter Notebook, Google Colab, Anaconda, VSCode.

Work Experience

Financial Relationship Banker (Huntington National Bank) May 2024 – December 2024

Technical Environment: AWS, SageMaker, OpenAI, Azure OpenAI, Postgres, Python (LlamaIndex, LangChain, Pandas, NumPy, chromadb, psycopg2), RAGAS framework, PyPDF2, python-pptx.

- Guided senior citizens, recent immigrants, and first-time banking customers through digital banking adoption, including mobile check deposit, secure password setup, and phishing/fraud recognition
- Analyzed client portfolio and product adoption data to support branch-level KPI reporting and retention strategy
- Contributed to a 30% improvement in client digital engagement by translating technical banking tools into plain-language guidance
- Maintained 100% audit compliance across high-volume datasets through rigorous data validation

Process Executive (Cognizant Technology Solution) August 2015 - February 2016(India)

- Managed large-scale data intake and processing, ensuring timely delivery of accurate output information to meet customer requirements.
- Collaborated with cross-functional teams during Agile ceremonies, including sprint planning, backlog grooming, daily stand-ups, sprint reviews, and retrospectives, to improve team productivity and deliverables.
- Partnered with Product Owners to identify, classify, sort, and route business-critical documents, ensuring quicker access to necessary details and facilitating efficient decision-making.
- Leveraged data analysis and documentation skills to streamline workflows, enhancing operational efficiency and accuracy.
- Contributed to the creation and improvement of business processes through detailed data management and analysis, supporting strategic initiatives and business goals.

Enterprise Summary

- Experienced with version control systems like GitHub, Bitbucket to manage the versions and configurations of the code.
- Excellent understanding of Agile and Scrum development methodology.
- Good experience in developing web applications implementing MVC architecture using Spring, REST.
- Experience in working with multiple vendors and leading geographically distributed teams.

Education

Master of Computer Application |Haryana Maharishi Markandeshwar University May 2015 • Focused on advanced computing concepts, data analytics, and system design.

- Equipped with strong programming skills and expertise in emerging technologies like AI, cloud computing, and cybersecurity.

Bachelor of Computer Application |Bihar Magadh University August 2011

- Emphasize fundamental Knowledge of Computer Science, Software Applications, web developer.

Certification

- AWS Certified Developer
- Full stack Web Developer Certification by Meta • Python for Data Science
- 3D Modeling by University of Michigan
- Better Scrum by Stay Amazing

Personal Project

Suitability & Product Recommendation Agent

Python, LangGraph, Streamlit, Gemini/LLM API

- Built a multi-agent decision-support tool that generates suitability-scored investment recommendations for financial advisors based on client profile (category, funds source, risk tolerance, time horizon, liquidity needs)
- Designed rule-based logic differentiating recommendations by client type (individual, senior citizen, trust, estate, business) with automatic tier classification (Retail through Private Client) and asset allocation modeling

- Deployed as a live Streamlit web app with real-time suitability scoring, risk-band placement, and compliance-flagged output for advisor review

Live demo: [<https://huntington-rag-agent-xifiwhyfnvmxpy2v99xdcp.streamlit.app>]

Marketing Content Clustering & Insights Pipeline

Python, Scikit-learn, TF-IDF, K-Means, PCA, SQL

- Built an end-to-end NLP pipeline processing 200 marketing content pieces across 6 channels (LinkedIn, Blog, Email, Instagram, Twitter, Webinar), transforming unstructured text into clustered, comparable content groups
- Applied TF-IDF vectorization to convert content into feature vectors, then used K-Means clustering to group content by topic and tone
- Used silhouette scoring across K=3 to K=8 to select the optimal cluster count, choosing K=8 based on the highest score (0.2499), producing coherent, well-separated content groups
- Reduced dimensionality via PCA to visualize the 8 clusters in 2D space, revealing distinct content archetypes such as Thought Leadership, Customer Success, Product Launch, and Educational Content
- Analyzed cluster-level performance, finding Thought Leadership content drove the highest average engagement (78.5) and LinkedIn was the top-performing channel (75.1 avg engagement) across the dataset
- Built an interactive dashboard with cluster filtering, engagement/click/conversion metrics, and a monthly trend view to support content strategy decisions

Fraud Detection & A/B Testing Dashboard

Python, Pandas, SQL, Tableau, A/B Testing, Statistical Analysis

- Built an end-to-end fraud detection pipeline processing transaction-level data across regions, transaction types (wire, ACH, card, check, online), and time periods to flag anomalous, high-risk activity
- Engineered a risk-scoring model (0–100 scale) segmenting transactions into High/Medium/Low risk bands, surfacing seasonal fraud patterns including a holiday-season spike in flagged activity
- Designed and ran an A/B test comparing a baseline rule-based engine against a new detection model, measuring precision, recall, and false positive rate; the new model improved precision by 17.5 points (61.4% → 78.9%) and cut false positives by roughly half, validated as statistically significant ($p < 0.05$)
- Built an interactive Tableau-style dashboard with drill-down filtering by region and transaction type, a monthly fraud-rate trend view, and a live-filterable table of flagged transactions ranked by risk score
- Applied SQL window functions and aggregations to compute region- and category-level fraud rates for stakeholder reporting

DietBot – AI-Powered Nutrition Chatbot

Tools/Technologies: AWS (Lambda, S3, API Gateway), Python, Machine Learning, 3D Modeling, Natural Language Processing (NLP), TensorFlow/PyTorch.

- Designed and developed an AI-powered chatbot, *DietBot*, to provide personalized diet recommendations based on user inputs like age, weight, and dietary goals.
- Integrated AWS services (Lambda for backend processing, S3 for data storage, API Gateway for secure API management) to ensure scalable, serverless deployment.

- Implemented NLP algorithms to understand and process user queries effectively, improving chatbot response accuracy by 20%.
 - Leveraged Kaggle datasets to train machine learning models for predicting nutritional requirements and meal suggestions.
 - Incorporated 3D models to visually represent food portions and nutritional information, enhancing user engagement.
 - Optimized performance by fine-tuning ML models, reducing API response times by 30% and increasing chatbot accuracy.
 - Deployed and monitored the application using AWS CloudWatch, ensuring high availability and reliability.
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